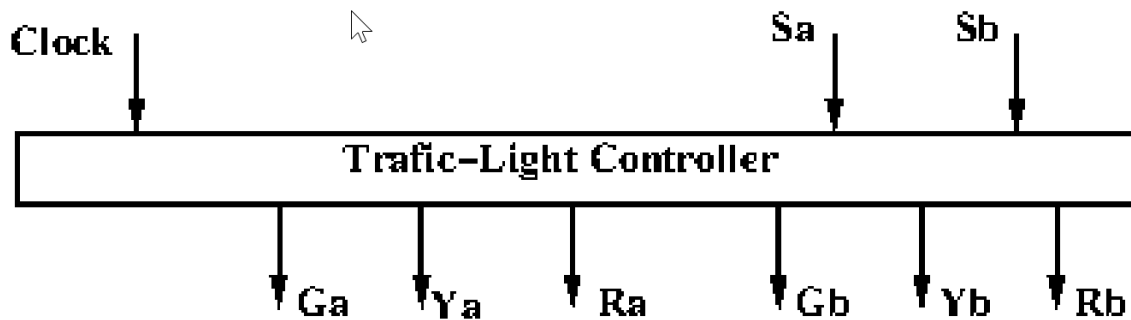


Problem 3



a) Design the traffic controller and show that the light controller is safe. Describe in detail why the design is safe.

I designed the controller with safety in mind in a few key ways. The first and most important design decision was to limit the states to 4 such that there only exists safe states, and all outputs are stored in a register. If by any series of events, the outputs become harmful, the traffic light enters the flashing red error mode and does not leave unless reset. Under no normal circumstances can the circuit enter an unsafe state, and if it does, it moves to the error state. The only issue with this design is that a transition between two safe states can be unsafe. If for example the yellow light stage is skipped, an accident could occur. While this is theoretically possible with my design, it would require a bit to randomly flip from state 1 to state 3 or vice versa.

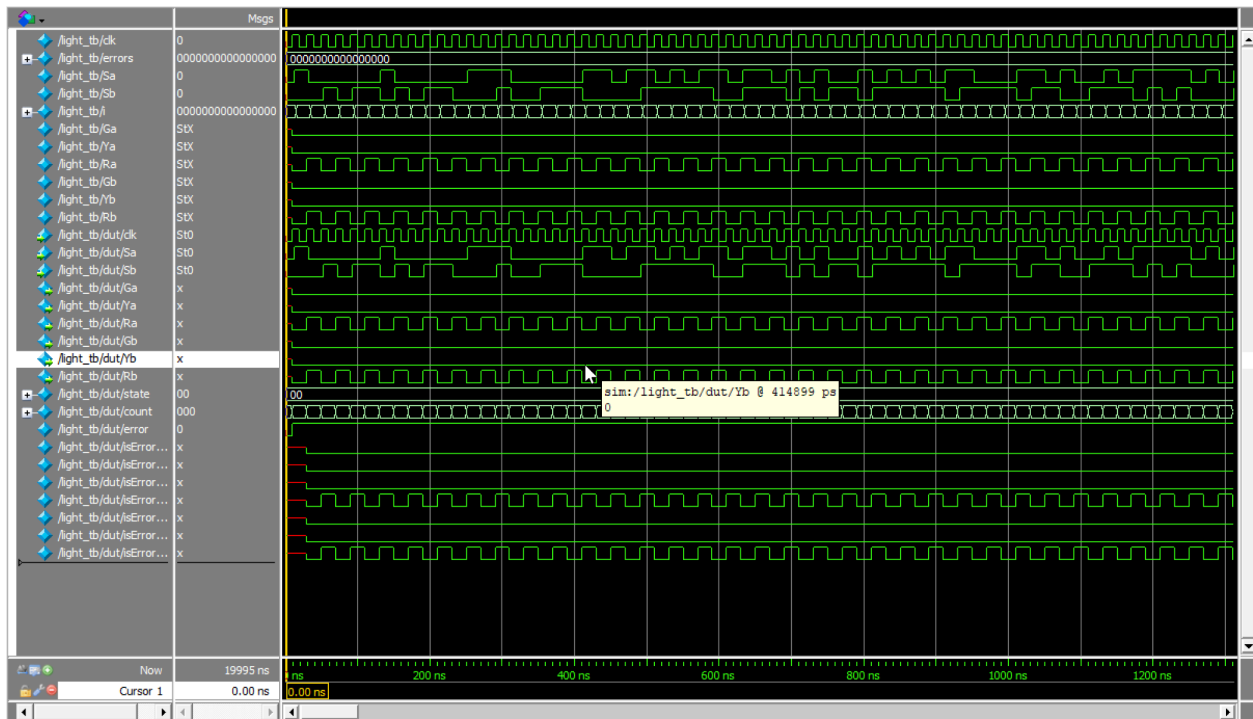
This could be fixed by tracking previous states. If the states do not follow a counting pattern (1,2,3,4), then the circuit could be put into error mode.

b) Write a Verilog model for the traffic light controller.

The model is written and called TrafficLightController.v

c) Simulate your model using ModelSim to verify correctness and safety of your design.

The model's waveform was viewed to confirm it conformed to all logic requirements. The model was tested against a testbench that looked for any unsafe outputs and it received zero errors. The model was also synthesized. All of this information is below:



```

=== Resource usage ===
Info (21058): Implemented 3 input pins
Info (21059): Implemented 6 output pins
Info (21061): Implemented 35 logic cells
Full report: C:\Users\luke\DocumentsReal\ECSE318\Github\ecse318\Labs\Lab2\P2\synth_TrafficLightController\TrafficLightController.map.rpt
PS C:\Users\luke\DocumentsReal\ECSE318\Github\ecse318\Labs\Lab2\P2>

```

Problems 5 Output Debug Console Terminal Ports

Reading pref.tcl

2020.1

```

# vsim -c -sv_seed 42 -do "run -all; quit -f" light_tb
# Start time: 12:25:55 on Oct 04,2026
# Loading work.light_tb
# Loading work.TrafficLightController
# run -all
# Errors: 0
# ** Note: $finish      : light_tb.v(31)
#   Time: 19995 ns   Iteration: 0   Instance: /light_tb
# End time: 12:25:55 on Oct 04,2026, Elapsed time: 0:00:00
# Errors: 0, Warnings: 0
PS C:\Users\luke\DocumentsReal\ECSE318\Github\ecse318\Labs\Lab2\P2>

```